

Communications issues in concurrent life-cycle design and construction

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Concurrent engineering offers the potential for a radical transformation of construction into a more competitive, more efficient and more collaborative industry. To take advantage of this potential, there is the need for effective communication at all levels within the industry. This paper explores the communications issues which are implicit in the adoption of concurrent engineering principles in construction. It identifies seven facets of communication which need to be addressed and highlights the need for efficient management of a collaborative communications infrastructure.

1. Introduction

The construction industry is well-known for its fragmentation. While this is more pronounced in the building sector than in the civil engineering sector, the key problems remain more or less the same. Each construction project involves several disciplines collaborating for a relatively short period in the development of the required facility [1, 2]. For a typical building project, participants would include the client, the architect, the structural engineer, the building services engineer, the quantity surveyor, the main contractor, sub-contractors, materials suppliers, etc. Many of these participants often work independently while taking decisions that inevitably affect others. Communication normally takes the form of 2D drawings, with periodic face-to-face project team meetings. In addition to the variety of professionals involved, the traditional approach to construction procurement favours a sequential approach to project development with, for example, the architect completing most of their design before the structural engineer who, in turn completes theirs before the services engineer, and so on. This 'over the wall' approach [3] to project development has resulted in numerous problems for the construction industry [4]:

- inadequate capture, analysis and prioritisation of client requirements,
- the fragmentation of design and construction information with data generated at one stage not automatically available for reuse 'downstream',
- poor integration, co-ordination and collaboration between the functional disciplines involved in a project,
- the lack of true life-cycle analysis of projects,
- the lack of communication of design intent and rationale,

- unwarranted design changes, disputes and liability claims,
- increase in design cost and time.

Efforts made towards process improvements in the construction industry have been based on the automation of various aspects of the design and construction process, and the use of relatively new procurement methods such as design and build. The absence of a co-ordinated and strategic approach to the development of the computer tools resulted in the so-called 'islands of automation', which reinforced rather than broke down traditional barriers [5]. Some measure of integration was achieved using the design and build (D&B) procurement method, but this has been limited by the continued use of discrete computer aided engineering (CAE) tools, the lack of effective integration of the project team, inadequate communications protocols, and the use of inappropriate management techniques. Other problems associated with the D&B method include poor design quality, lack of certainty of expected performance, high tendering costs, and lack of flexibility in accommodating client changes [6].

To address the problems of the construction industry effectively requires the adoption of concurrent engineering principles. This involves the integration of all project team members within a computer-integrated framework for concurrent life-cycle design and construction (CLDC). Such a framework is being developed in a collaborative project between the Universities of Teesside and Newcastle and is illustrated in Fig 1.

Within the framework, the integrated design and construction process is underpinned by a variety of design

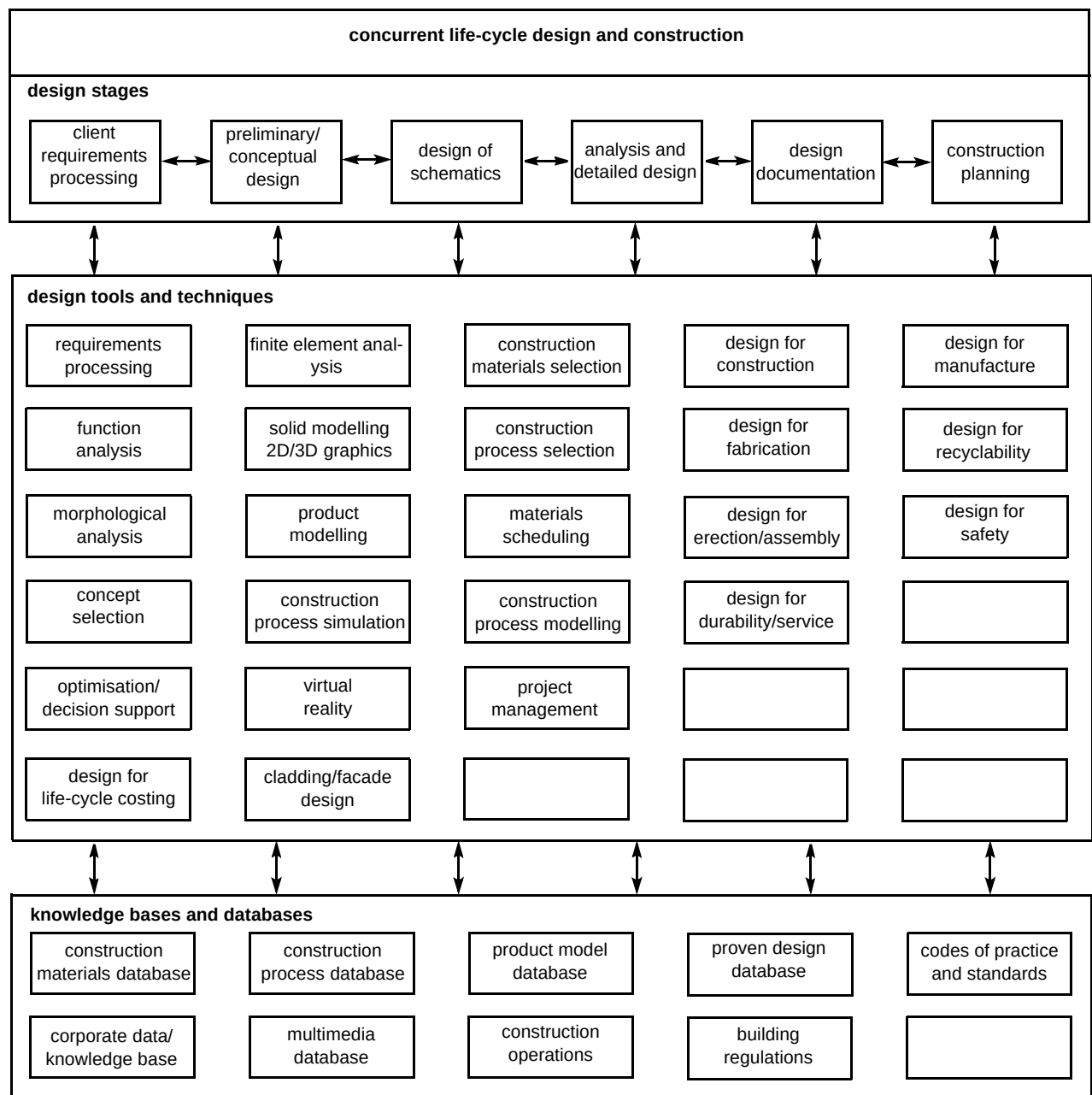


Fig 1 Integrated CLDC framework.

tools and techniques, and by appropriate databases and knowledge bases. An integral aspect of the overall framework is ensuring effective communication of design information at all stages of the integrated design and construction process. The key communications issues which need to be addressed are discussed below.

2. Communications issues

2.1 Considerations

There are several communications issues inherent in the adoption of concurrent engineering principles in the construction industry. These have been identified taking into account the following considerations which have a bearing on the desired communications infrastructure:

- concurrency in an integrated design and construction process requires greater discipline in the production, manipulation, storage and communication of design information,
- design information necessarily consists of both graphical and non-graphical information which must be communicated between members of the project team,
- the greater the level of concurrency in a process, the greater the level of co-ordination required — this entails an increased level of communication between the various stages and activities in the process, as well as between the project team members,

- paper-based communication of design information is now inadequate to cope with the high level of functionality (in terms of speed, accuracy, usability, ease of modification, enhanced visualisation, improved co-ordination, etc) required in a collaborative design environment,
- the increasing ‘globalisation’ and complexity of construction projects means that project teams often involve partners from widely distributed geographical areas, sometimes on different continents [7] — as face-to-face meetings in such circumstances are expensive in terms of time, money and personal inconvenience [8], effective communications protocols able to collapse time and distance constraints are therefore necessary,
- the very fast pace of technological development, particularly in computing and telecommunications dictate that, for the construction industry to remain competitive, it must take advantage of new and emerging technologies such as multimedia, virtual reality, broadband communications networks, etc.

2.2 Facets of communication

There are seven facets of communication (illustrated in Fig 2) which need to be addressed in concurrent life-cycle design and construction (CLDC):

- communication between intra-disciplinary CAE tools (level 1),
- communication between each project team member and their design tools (level 2),
- communication between design team members (level 3),
- communication between each discipline and the common project model (level 4),
- communication across the stages in the project life-cycle (level 5),
- communication between the project team and third parties (level 6),
- communication between inter-disciplinary CAE tools (level 7).

Intra-disciplinary tool-to-tool communication

This refers to the communication between the various design tools within each discipline. For example, the structural engineer’s finite element (FE) analysis model needs to be able to communicate with the elemental design and detailing packages or with a knowledge-based conceptual design system. This facet of communication is

essential to integration within a given discipline and will, in addition to minimising data input and rework, ensure that all design information emanating from a given discipline is mutually consistent.

Designer-to-tool communication

It is important that designers are able to communicate effectively with the design tools at their disposal. This level of communication is often a function of the user-interface design, and is necessary for the effective deployment of the tools. In this regard, it is essential that CAE tools are designer-oriented [9], providing the designer with the flexibility to structure design information in accordance with individual, project and/or corporate requirements, and to configure the user interface to suit individual preferences. This will entail taking advantage of emerging technologies such as multimedia, voice/motion recognition, and virtual reality.

Project team communication

This relates to communication between the various members of a given project team. There is a need to provide for an appropriate communications infrastructure which will facilitate multilateral communication involving all members of the project team or a subset thereof. Concurrent project development requires that all appropriate team members participate in meetings, agreeing the basis for design decisions, and resolving potential downstream safety, buildability or other problems. This level of communication should provide support for both synchronous and asynchronous ‘meetings’, as well as both co-located working (such as within a multi-disciplinary practice) and distributed working (within a single organisation, and/or between an extended network of partners). In addition, enhanced visualisation tools based on the ‘What You See Is What I See’ (WYSIWIS) philosophy (but also allowing for multiple views of design information [10]) are necessary to support concurrent project development.

Discipline-to-project model communication

At this level, each discipline should be able to communicate interactively with a common project model. A common project model is considered vital for ‘seamless interworking’ between several disciplines. Each discipline should, therefore, have facilities to insert and abstract information from the project model, as well as view (and comment on) changes to the model instigated by other disciplines. It is also essential that, where appropriate, intra-discipline design tools are compatible with the hardware and software platforms on which the common project model is based. This will facilitate bi-directional co-ordination. The common project model needs to support concurrent

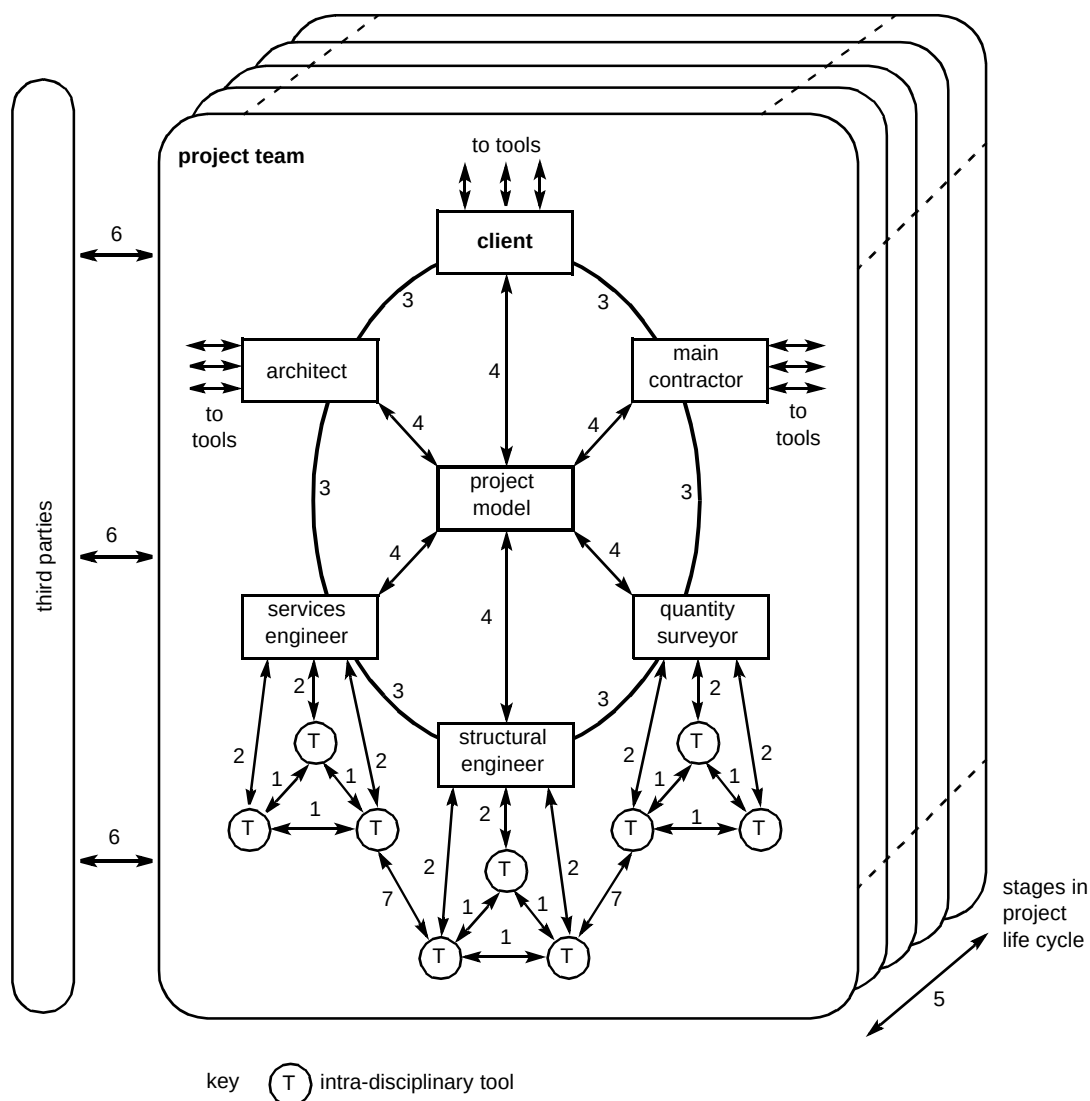


Fig 2 Facets of communication in CLDC.

multi-user access, and provide appropriate mechanisms for integrity and consistency checking.

Communication between stages in the project life cycle

The integrated framework for concurrent life-cycle design and construction provides for six key design stages — client requirements processing, preliminary/conceptual design, design of schematics, analysis and detailed design, design documentation, and construction planning. There is need for communication of both design information and design rationale between these stages as well as the subsequent stages of construction, utilisation, maintenance, and disposal (Fig 3). It should be borne in mind that the configuration of the communication facets shown in Fig 2 will change from one stage of the project life cycle to another. In other words, not all the facets will be 'active' at each stage; this is discussed in greater detail elsewhere [11].

project life cycle will not only facilitate the re-use of design information at the later stages of a project's life cycle, but also ensure the traceability of design rationale and decisions to explicit and implicit client requirements. This can be provided for within the common project model, and will prevent ill-advised late changes and limit disputes and claims.

Project team-to-third party communication

In construction, third parties are often introduced into a project and, while they are not full-time project team members *per se*, there is a need for adequate communications between the project team and such parties. Third parties may include specialists (e.g. geotechnics experts), regulatory bodies (e.g. local authority building inspectors, Health & Safety Executive safety inspectors) or others. The involvement of some third parties, although transient, may involve a high level of communication with

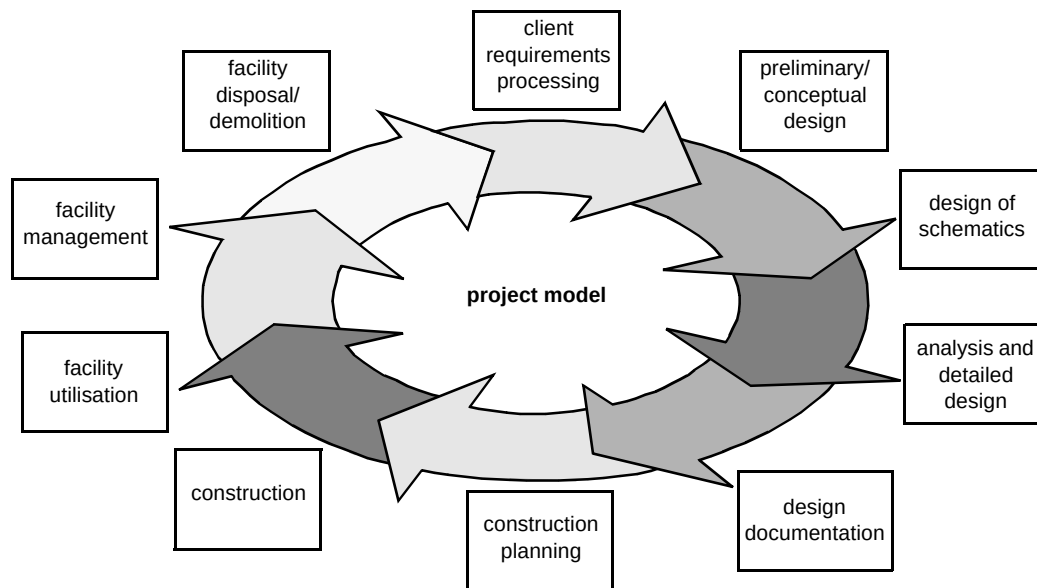


Fig 3 Communication across stages in project life cycle.

the project team. In such cases, it may be necessary to allow third parties limited access to the project model (or parts thereof) and/or the project team's communications network.

Inter-disciplinary tool-to-tool communication

This is the communication between the CAE tools used by the different disciplines. For example, there may be a need for the structural engineer's CAE system to exchange information directly with the architect's CAD modelling system, or the services engineer's design system. This level of communication is sometimes useful and can readily be undertaken where the relevant disciplines utilise the same or mutually compatible systems. However, where disparate systems are in use, necessitating numerous interfaces, this level of communication would be best carried out via the common project model.

3. Communications management

In addressing all the above facets of communication, there are a number of managerial issues which also need to be addressed. Some of these include access control, version control, change propagation and management, data integrity and security, communication etiquette, and information flow management.

3.1 Access control

It is essential that an elegant solution is provided for the problem of access control. This should assign and enforce access rights to the common project model for all members of the project team. Read access to the project model should be available to all appropriate project team members while right-to-modify access should be retained by the creator of

any given aspect of the model. Access to the project team's communications infrastructure also needs to be controlled but project team members should have ready access to third parties and information repositories such as the Internet.

3.2 Version control

Engineering design is by nature an iterative process. This is even more so in a concurrent engineering environment which is known to generate more early design changes than the conventional approach, while reducing late (and expensive) design changes to the barest minimum. Thus, an effective version control system is necessary. Design team members need to communicate on the basis of the most up-to-date version of the project model but also require the flexibility to refer to previous or alternative versions. A version management system needs to cover the project model itself, as well as the underlying design intent and rationale. It should also be possible for each discipline to retain 'development versions' of their own part of the project model, and to commit these to the common model only at appropriate stages, such as when proposed changes have been negotiated and agreed.

3.3 Change management

Another important aspect of communication in a concurrent engineering environment is the management of design changes. Clear protocols need to be established for change notification, propagation and management. The guiding principles should include the following:

- all parties affected by a given change should be notified promptly, with an indication of how the changes might affect them,
- changes should be highlighted in the project model for all project members to see,
- time should be allowed for negotiation of changes prior to permanent committal of the changes to the common project model,
- the automatic propagation of changes to all appropriate parts of the project model should take place only after the proposed changes have been agreed by all relevant parties,
- the rationale for all significant changes should be recorded to facilitate traceability of design intent and rationale, and prevent uninformed (and possibly contradictory) changes at later stages of the project life cycle.

3.4 Data integrity and security

Data integrity and security are closely related to the aforementioned issues of access control, version control and change management. Mechanisms need to be set up to prevent data held in the project model from being corrupted or abused by parties external to the project team. Efficient procedures for back-up will also be necessary to minimise problems due to accidental loss or corruption of project data. In addition, it may be important to protect the team (and/or individual team members) by restricting access to their contact details.

3.5 Communication etiquette

An etiquette for communication between project team members is important for ensuring smooth collaboration. Group dynamics are influenced to a large extent by the personalities of the individuals involved. The agreement of an appropriate code of practice will limit anti-social behaviour and ensure that cultural and social conventions are not breached. This is very important when the collaborating parties are in culturally diverse geographical locations. Such a code of practice should cover all forms of communication used by the team (e.g. oral, written, video, etc) and be applicable to both synchronous (or simulated meeting room) communication and asynchronous communication. It should be recognised that the group dynamics of project teams often differ, and what works for one team may be completely unacceptable to another team with a slightly different composition.

3.6 Information flow management

Effective communication management in a concurrent engineering setting entails efficient information flow management. The strategy should be to ensure that all stages of the design and construction process (and the activities implicit in them), as well as the members of the project team, have the right information, of the right quality, at the right time, and at the right level of granularity. Means of dealing with (detecting and resolving) information bottlenecks and overload have to be devised. In this regard, it is vital that the dependencies of activities are well understood, and an information flow monitoring system set up.

4. Discussion

Clearly, the construction industry needs to adopt concurrent engineering techniques in order to effect a positive change in its culture and business processes. The communication issues raised in this paper all need to be addressed to ensure that the full benefits of concurrent engineering are realised. A proactive approach by all parties to a construction project is essential. Clients in particular, as the financiers (or promoters) of construction projects, are uniquely placed to champion the course of change. It is encouraging that a number of client organisations are already exploring ways of improving communication and co-ordination on their projects through maintaining close partnerships with both consultants and contractors. In some cases, this has involved insisting that all members of the project team use the same CAE system. While there are obvious advantages in this approach, there are also fundamental drawbacks which make the case for a global solution that enables the various disciplines to use their own CAE systems, and to contribute to a common project model.

Bowles [12] identifies the primary objectives for IT and communications to support concurrent engineering as:

- to reduce the effect of distance so team members can interact as if co-located,
- to enable cost-effective, flexible applications to have a visual object representation,
- to manage the generation, storage and distribution of data,
- to facilitate the integration of applications so the man/machine interface solves the business problem and not vagaries of implementation — in particular, to provide smooth transfer between design, modelling, test and production.

Providing the facets of communication described in this paper will enable these primary objectives to be addressed in concurrent life-cycle design and construction (CLDC). Secondary benefits in terms of productivity gains, quality improvements, time savings, cost savings, greater client satisfaction, improved project team dynamics and *esprit de corps*, etc, will also be realisable. Additionally, information will be freely available in an effective social environment [12].

The identified communication facets address the three main modes of communication illustrated in Fig 4 — person-to-person, person-to-computer, and computer-to-computer.

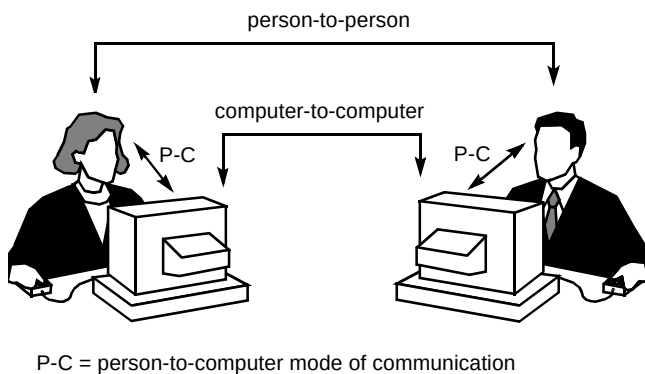


Fig 4 Modes of communication.

The construction industry has traditionally relied on face-to-face (FTF) meetings, telephony, and printed 2D drawings (delivered by hand or by post) to satisfy the requirements of person-to-person communication of design and other project information. More recently, there has been a tremendous growth in person-to-computer communication, as the industry has moved to extend the application of computer-aided engineering (CAE) beyond analysis to drafting, detailed design, conceptual design, project planning and project management. This has resulted in numerous benefits for those firms that have taken the plunge, and invested in CAE. Unfortunately, very little progress has been made in computer-to-computer communication, with many firms (that use the same hardware and software) still sending floppy disks by post.

Effective communication within a concurrent life-cycle design and construction environment requires that adequate provision is made for all three modes of communication highlighted above. The seven facets of communication discussed earlier in the paper can be categorised into these modes of communication, and provide a coherent framework for ensuring that an adequate communications infrastructure is established. It is clear from the foregoing that existing procedures adopted within the construction industry are grossly inadequate to support concurrent project development. There is major scope for improvement

based on the use of new and emerging information and communication technologies.

5. Summary and conclusions

Good communication is vital for the success of concurrent engineering in construction. This paper has sought to identify the key communication issues which need to be addressed. A multi-faceted approach is advocated which will provide support for communication between intra- and inter-disciplinary design tools, project team members and their intra-discipline tools, project team members, different disciplines and the common project model, the various stages in the project's life cycle, and between the project team and third parties. The management of the communications infrastructure, which addresses all the above facets of communication, is also discussed and the importance of effective controls at all stages is emphasised. Effective implementation of all the levels of communication highlighted will go a long way towards ensuring the success of concurrent engineering in a highly fragmented construction industry. Some of the enabling information and communication technologies which have a role in facilitating this are already in existence, while others are being developed as part of a European project on Collaborative Integrated Communications in Construction (CICC). These technologies are reviewed in detail elsewhere [13].

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